

MetricVault

Subscription Revenue Data Product

dbt + MetricFlow + LookML + Dagster + Snowflake + Tableau, with a real SOX reconciliation control

Every pipeline stage in this deck actually ran. dbt build: 40/40 tests passing. Dagster: full asset graph, RUN_SUCCESS.

CONTEXT

The Problem

- A subscription company's revenue metrics get defined multiple times: a BI calculated field, an analyst's ad hoc SQL, Finance's spreadsheet, and they quietly drift apart.
- Because the company is public, the numbers this pipeline produces directly support financial reporting, so accuracy, lineage, and Finance reconciliation aren't optional.
- The role: own the shared transformation and metric layer everything else depends on, not a pipeline role, not a dashboard role.

Where the Data Comes From

- Synthetic data shaped to mirror real Stripe API objects exactly: Customer, Subscription, Invoice, Charge, Refund, plan-change events.
- 18-month history, documented growth/churn model: 4,503 customers, ~27,000 invoices, ~3.2% monthly churn.
- Finance's independently-recorded ledger (NetSuite GL stand-in) is generated separately, with deterministic noise, so reconciliation results are real and reproducible.

HYPOTHESIS

What We're Testing For

- Whether MRR movement classification, the genuinely hard part of subscription analytics, can be built as governed dbt logic both MetricFlow and LookML read identically.
- Whether an independent Finance reconciliation catches real variance rather than trivially always passing.
- It does: 2 of 18 months fail the 2% control threshold, tied to a documented one-time adjustment, not silently absorbed.

Certified Metrics, One Definition Each

- Primary: dbt Semantic Layer / MetricFlow, ARR, MRR, NRR, Logo Retention, Paying Customers, and every MRR movement type, each with exactly one definition.
- Secondary: LookML, a migration-period mirror for existing Looker consumers, wired to the same certified mart tables, never independently defined.
- Verified: semantic_layer_refresh validated 9 metrics against their underlying measures, live, in the Dagster run.

Stack & Why It Fits

Layer	Tool	Why
Ingestion	Stripe-shaped extract + Dagster	Matches target ingestion + orchestration stack
Warehouse	Snowflake (verified via DuckDB)	Runnable and testable without a live account
Transformation	dbt (staging → intermediate → marts)	Version control, tests, CI, documentation
Semantic layer	MetricFlow (primary) + LookML (migration)	Certified metrics, one source of truth
Orchestration	Dagster	Real asset graph + SOX asset check
BI	Tableau	Executive ARR/MRR bridge, retention, reconciliation

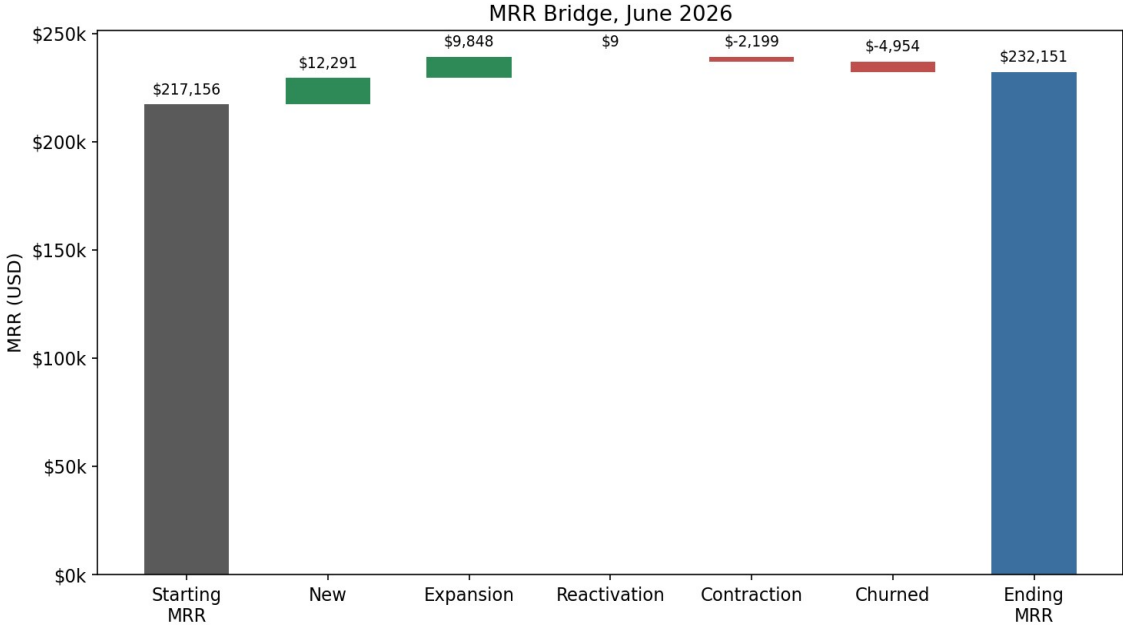
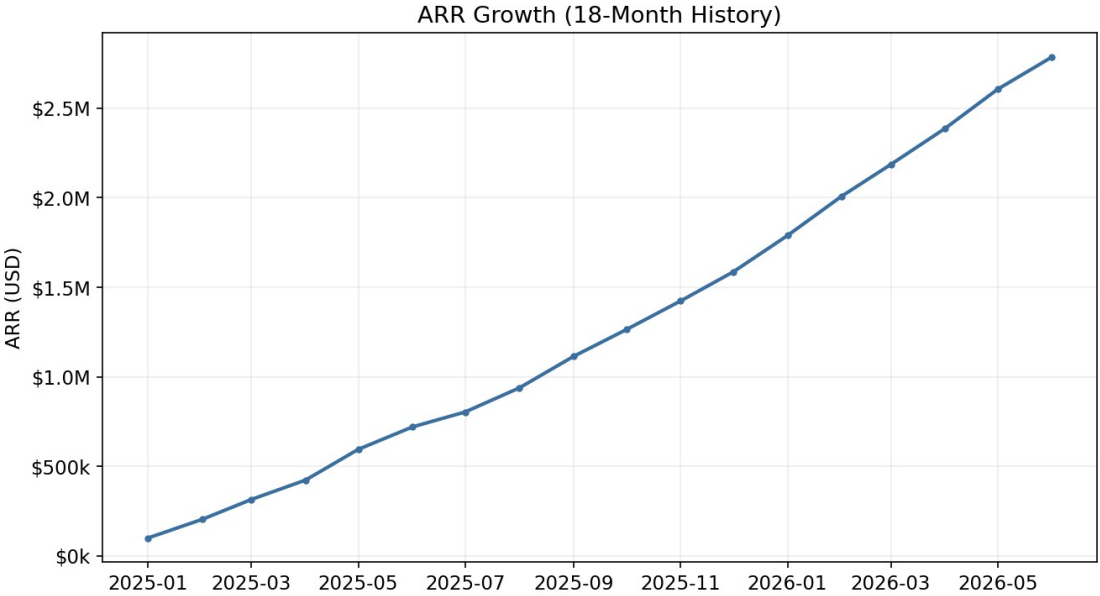
COMPLIANCE

The SOX Control, Live

- `fct_finance_reconciliation` compares data-team ARR against Finance's independently-recorded ledger, every month, automatically.
- The Dagster asset check ran for real and correctly flagged two months by name and exact variance, without blocking the pipeline.
- Access control, change management, and this reconciliation are each documented and mapped to a specific SOX ITGC control objective in `sox/`.

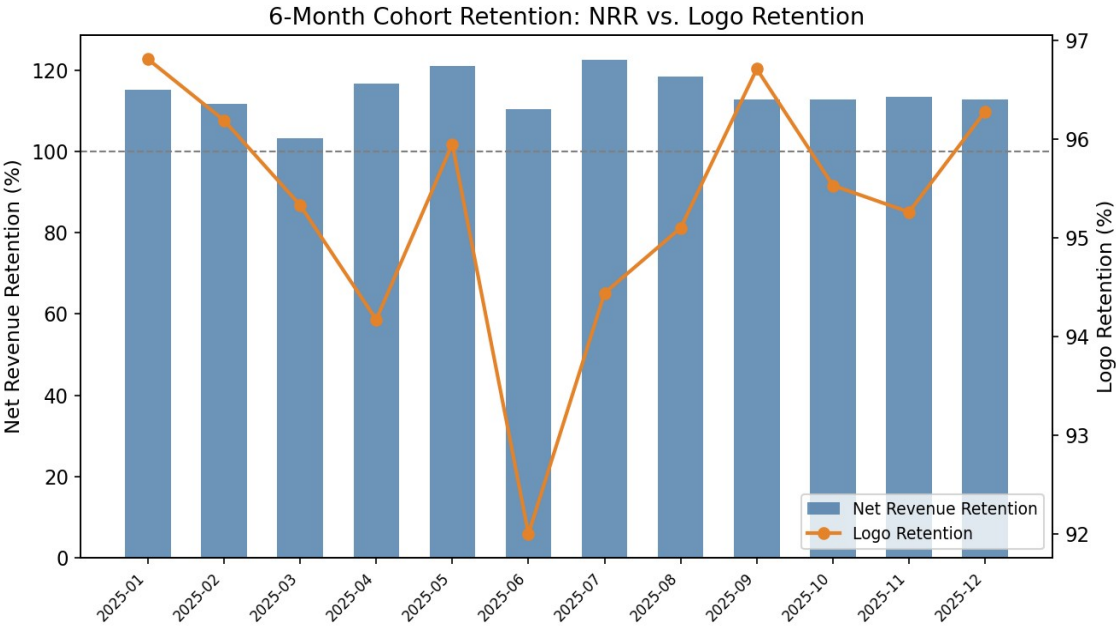
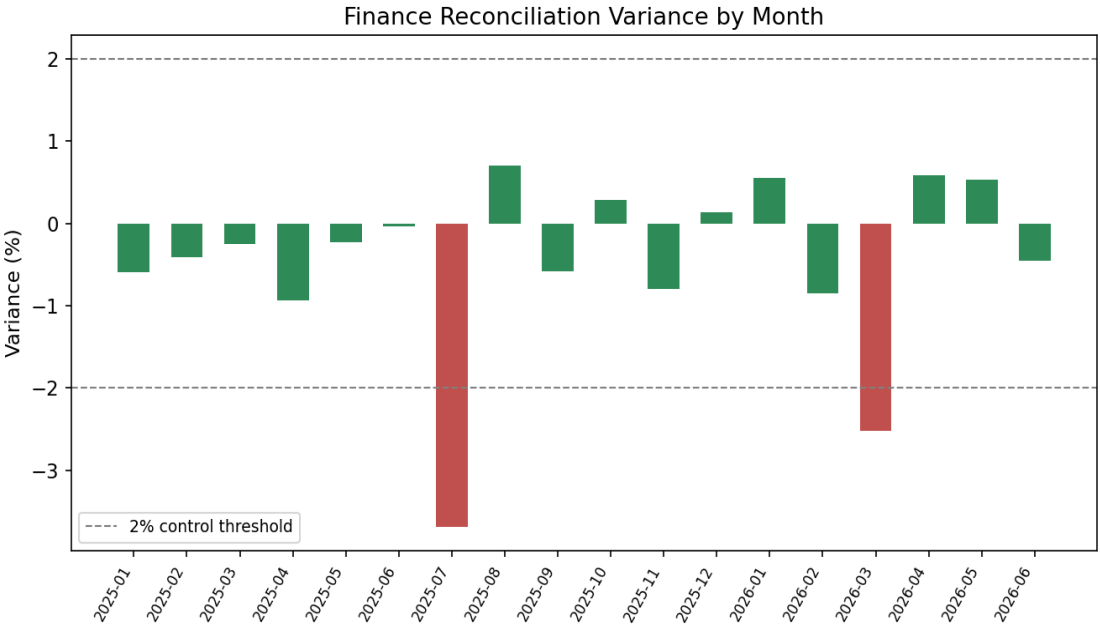
KPI SNAPSHOT

Results



- ARR grew from ~\$100K to \$2.79M over 18 months, 2,888 active paying customers.
- 114.3% average Net Revenue Retention, 95.3% average Logo Retention across 6-month cohorts.
- 40/40 dbt tests passing. Dagster: full asset graph RUN_SUCCESS.

Reconciliation & Retention



16 of 18 months reconcile within the 2% control threshold; the two flagged months (Jul 2025, Mar 2026) trace to a documented one-time GL adjustment, exactly what this control is designed to catch.